

COLVILLE LAKE, NT, Canada

WMO: 710550

Lat: 67.0214N

Lon: 126.1231W

Elev: 850

StdP: 14.25

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.1	-39.1	-50.8	0.3	-42.5	-46.8	0.4	-38.8	21.7	-3.5	19.7	-1.9	2.0	200	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	75.7	59.9	71.9	58.6	68.3	57.1	62.5	72.2	60.3	69.0	58.3	66.4	9.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.5	75.6	65.9	56.4	69.8	64.4	54.1	64.3	62.5	28.6	72.2	27.0	69.4	25.7	66.5	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.3	18.7	16.4	-47.5	82.7	5.4	3.3	-51.4	85.1	-54.6	87.0	-57.6	88.9	-61.6	91.3		
			-46.8	64.8	4.4	3.1	-50.0	67.0	-52.6	68.8	-55.1	70.5	-58.3	72.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	-15.4	-12.3	-5.8	13.9	35.6	51.6	58.0	52.0	40.2	22.9	1.6	-10.0	(6)
	DBStd	28.61	14.40	13.44	13.55	12.74	10.35	8.21	7.18	7.59	7.87	8.61	11.96	13.39	(7)
	HDD50	11661	2028	1744	1729	1084	457	80	16	68	304	840	1451	1860	(8)
	HDD65	16623	2493	2164	2194	1534	912	406	236	409	744	1305	1901	2325	(9)
	CDD50	539	0	0	0	0	10	127	263	128	11	0	0	0	(10)
	CDD65	25	0	0	0	0	0	3	18	4	0	0	0	0	(11)
	CDH74	195	0	0	0	0	1	44	127	23	0	0	0	0	(12)
	CDH80	29	0	0	0	0	0	6	21	2	0	0	0	0	(13)
	WSAvg	7.1	5.3	5.6	5.9	6.7	7.6	8.3	8.8	9.1	8.9	7.8	6.2	5.2	(14)
	PrecAvg	12.9	0.7	0.8	0.6	0.5	0.8	1.5	2.0	2.0	1.5	1.0	0.9	0.7	(15)
(16) Precipitation	PrecMax	19.8	1.5	4.7	1.3	1.5	2.4	2.8	3.5	3.2	3.0	1.6	1.9	1.4	(16)
	PrecMin	9.4	0.1	0.2	0.1	0.0	0.1	0.4	0.8	0.9	0.9	0.5	0.4	0.2	(17)
	PrecStd	2.2	0.4	0.8	0.3	0.4	0.4	0.6	0.7	0.7	0.5	0.3	0.4	0.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	24.0	33.9	49.0	69.5	79.4	83.0	76.9	63.8	46.8	31.7	22.7	(19)
		MCWB	26.3	22.5	30.4	40.5	52.7	59.6	63.8	61.5	52.6	41.5	31.2	22.0	(20)
	2%	DB	18.5	16.9	25.1	41.9	62.5	73.5	77.4	71.7	58.5	38.8	25.2	17.1	(21)
		MCWB	17.7	15.9	23.0	36.2	49.3	57.2	61.2	60.3	50.1	36.8	24.5	16.5	(22)
	5%	DB	10.0	11.4	19.0	37.3	56.4	69.3	73.4	67.4	54.8	35.0	21.7	13.3	(23)
		MCWB	9.4	10.7	17.6	33.1	45.6	55.0	59.5	58.1	48.3	33.4	21.0	12.7	(24)
	10%	DB	3.8	6.6	13.6	33.1	51.1	65.3	69.9	63.5	51.3	32.8	18.2	9.5	(25)
		MCWB	3.2	5.9	12.3	30.1	42.8	53.5	58.5	56.0	46.8	31.4	17.5	8.9	(26)
	0.4%	WB	26.2	22.5	30.5	41.0	53.0	61.2	66.3	63.9	54.7	42.0	31.4	22.2	(27)
		MCDB	27.9	23.8	33.5	48.7	68.1	74.6	77.3	73.6	59.9	45.3	31.9	22.8	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	17.8	15.9	23.0	36.6	49.5	58.7	63.5	61.4	51.5	36.7	24.7	16.5	(29)
		MCDB	18.5	16.8	24.9	41.6	61.4	70.0	73.8	69.3	56.8	38.6	25.2	16.9	(30)
	5%	WB	9.5	10.7	17.5	33.4	46.4	56.6	61.6	58.8	49.1	33.6	21.1	12.7	(31)
		MCDB	10.0	11.4	18.9	36.7	55.6	67.1	70.8	66.3	53.7	34.8	21.7	13.1	(32)
	10%	WB	3.2	5.9	12.3	30.1	42.9	54.3	59.7	56.6	46.9	31.6	17.6	8.9	(33)
		MCDB	3.8	6.6	13.5	33.3	50.9	63.7	67.9	63.0	51.2	32.7	18.1	9.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	14.9	18.1	20.5	17.3	17.3	14.9	12.9	10.4	8.4	12.3	12.8	(35)
		MCDBR	18.9	17.8	22.7	21.9	24.9	23.5	20.3	18.7	16.7	9.2	12.7	15.3	(36)
		MCWBR	18.5	17.2	20.2	16.7	14.5	11.6	9.5	10.0	10.0	7.5	12.4	14.9	(37)
	5% WB	MCDBR	19.0	17.6	22.4	20.2	24.5	21.3	17.6	17.4	15.2	8.6	12.5	15.1	(38)
		MCWBR	18.6	17.0	20.1	15.4	14.7	11.4	9.2	9.9	9.8	7.3	12.2	14.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.148	0.208	0.252	0.293	0.331	0.338	0.355	0.331	0.291	0.236	0.153	N/A	(40)
	taud		1.932	2.218	2.231	2.200	2.340	2.451	2.426	2.505	2.563	2.284	1.935	N/A	(41)
	Ebn at Noon		83	217	259	274	274	276	266	262	250	196	70	0	(42)
	Edh at Noon		4	14	24	33	33	30	30	25	18	13	3	0	(43)
(44) All-Sky Solar Radiation	RadAvg		22	210	760	1492	1854	1946	1646	1150	627	197	42	0	(44)
	RadStd		2	17	33	68	141	150	123	106	61	21	5	0	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.60	+3.73	+3.87	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	+1.26	+2.50	+2.39	N/A	N/A	N/A	-396	-365	N/A	N/A	

Nomenclature: See separate page